Algorithms

Divide and Conquer

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Dynamic Programming

- Solve optimization problems using **overlapping subproblems**.
- Store solved subproblems to avoid recomputation.
- Typical gain: exponential recursion $\rightarrow$ polynomial/linear time.

When DP Applies + Design Pattern

Two required properties

- **Optimal substructure:** optimal solution is built from optimal subproblems.
- **Overlapping subproblems:** same subproblems repeat many times.

DP recipe

1. Define a state (what does DP entry represent?).
2. Write a recurrence relation.
3. Set base cases.
4. Compute with memoization (top-down) or tabulation (bottom-up).

Note: If subproblems do not overlap, Divide & Conquer is usually better.

Fibonacci: Why DP Helps

Naive recursion:

$$\mathsf{Fib}(n) = \mathsf{Fib}(n-1) + \mathsf{Fib}(n-2), \quad T(n) = O(2^n)$$

Repeated calls to the same subproblems cause exponential blow-up.

DP improvement

- **Top-down (memoization):** cache each $\mathsf{Fib}(i)$ once.
- **Bottom-up (tabulation):** fill $\mathrm{DP}[0 \ldots n]$ iteratively.

Both give $O(n)$ **time**. Space is $O(n)$ (or $O(1)$ with rolling variables).

Example 1: Coin Change (Min Coins)

Problem: coins $[c_1, \ldots, c_k]$, target n. Find minimum coins.

State: $\mathrm{DP}[i]$ = minimum coins to make amount i.

Recurrence:

$$\mathrm{DP}[i] = \min_{c \in \text{coins},\ c \le i} (\mathrm{DP}[i-c] + 1), \quad \mathrm{DP}[0] = 0.$$

Complexity: $O(nk)$ time, $O(n)$ space.

For coins $[1, 2, 5]$, target 11: answer is **3** $(5 + 5 + 1)$.

Example 2: LCS (Longest Common Subsequence)

State: $\mathsf{LCS}(i, j) = $ LCS length of prefixes $X[1..i]$, $Y[1..j]$.

Recurrence:

$$\mathsf{LCS}(i,j) = \begin{cases} 1 + \mathsf{LCS}(i-1, j-1), & X[i] = Y[j], \\ \max\{\mathsf{LCS}(i-1, j), \mathsf{LCS}(i, j-1)\}, & X[i] \neq Y[j]. \end{cases}$$

Base cases: $\mathsf{LCS}(0, j) = \mathsf{LCS}(i, 0) = 0$.

DP table size is $(m+1) \times (n+1)$, so $O(mn)$ **time** and $O(mn)$ space.

LCS Example: Final DP Table

Example: $X = \texttt{ABCB}$, $Y = \texttt{BDCAB}$

	ε	B	D	C	A	B
ε	0	0	0	0	0	0
A	0	0	0	0	1	1
B	0	1	1	1	1	2
C	0	1	1	2	2	2
B	0	1	1	2	2	**3**

Answer: LCS length is **3**. One LCS is BCB.

Example: The Gold-Extractor Game

Problem: Coins are hidden in an $n \times n$ integer grid.

- A player controls two extractors:
  - **Vertical** extractor (red) moves right by one column line.
  - **Horizontal** extractor (blue) moves up by one row line.
- At each step, move exactly one extractor by one line.
- If the new line contains coins, collect exactly one; all other coins on that same line are lost.
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Goal

Choose *which* extractor to move at each step to maximize total coins collected.

A Combinatorial Game

- Goal is to maximize the profit by carefully choosing which machine to move at each step.

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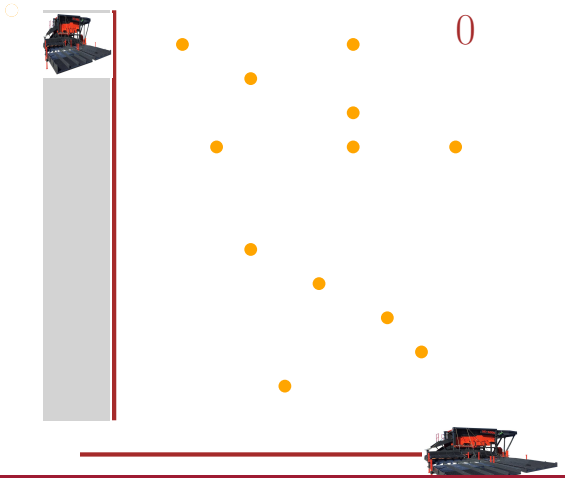

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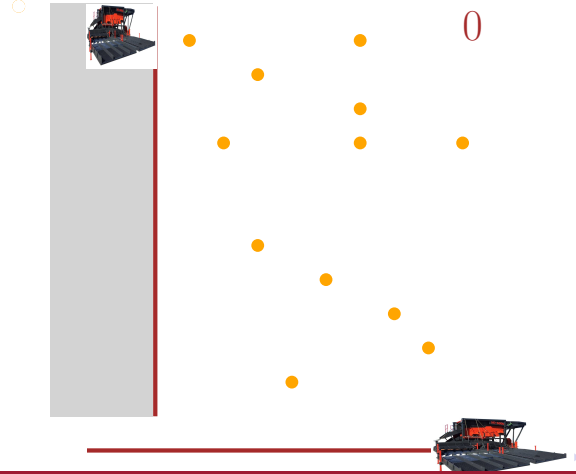

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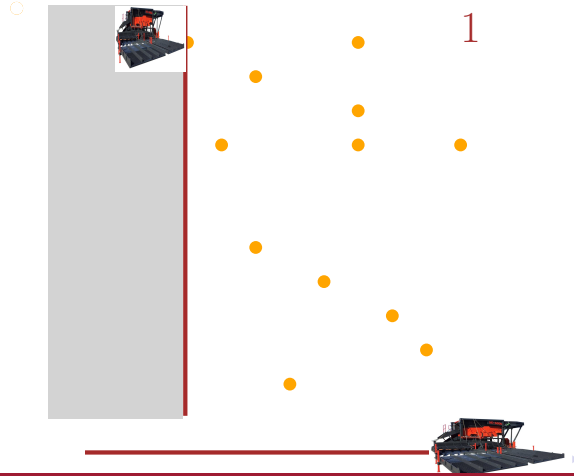

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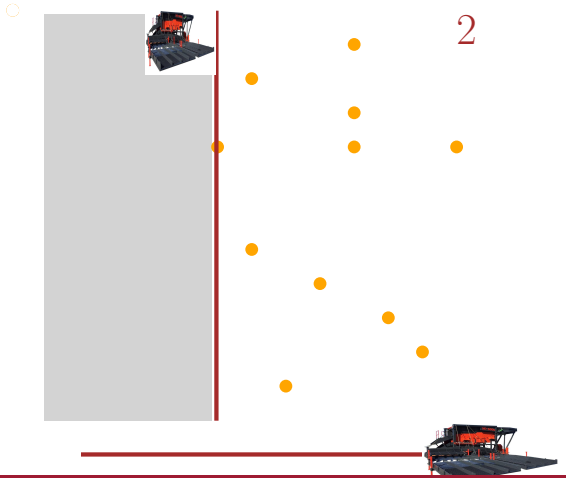

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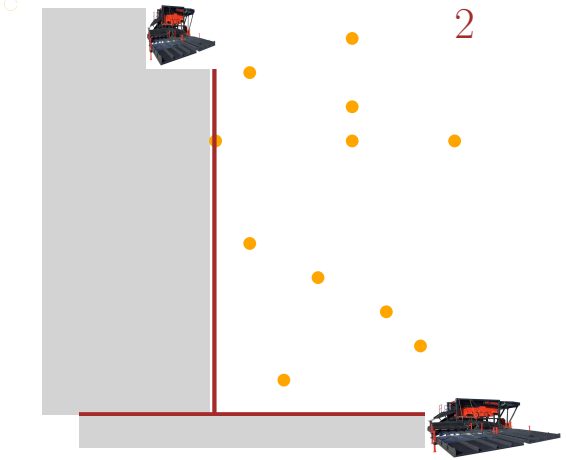

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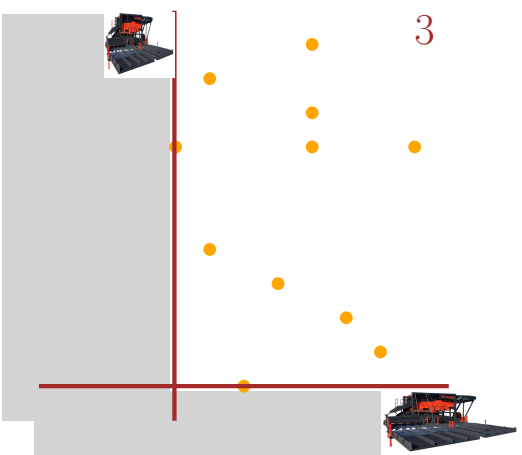

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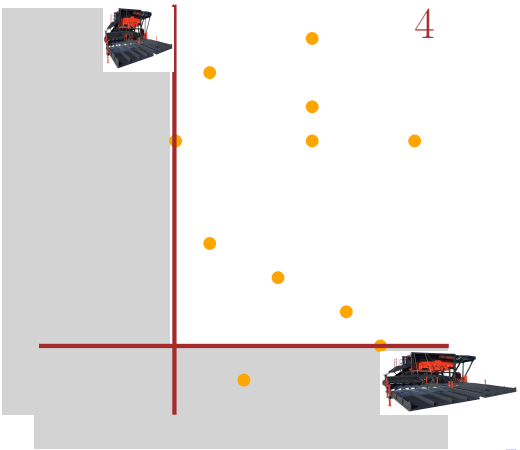

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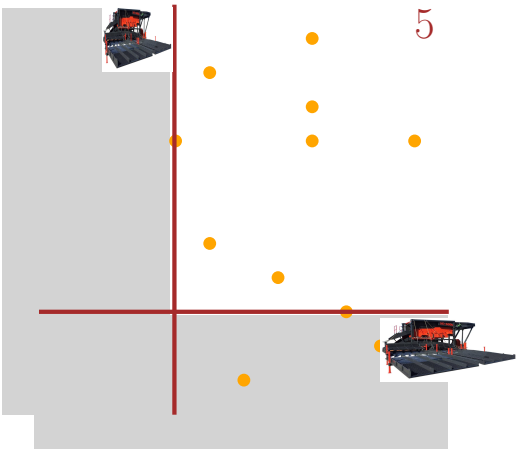

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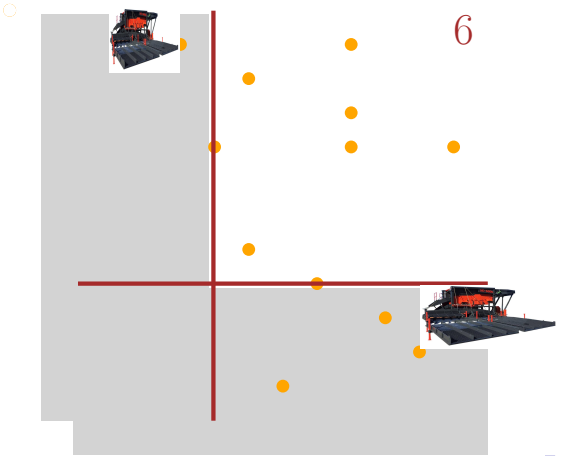

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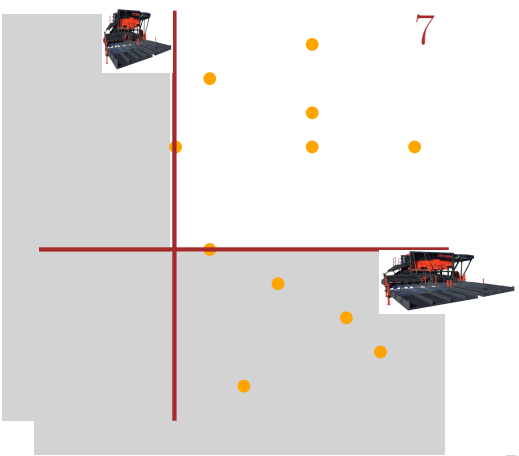

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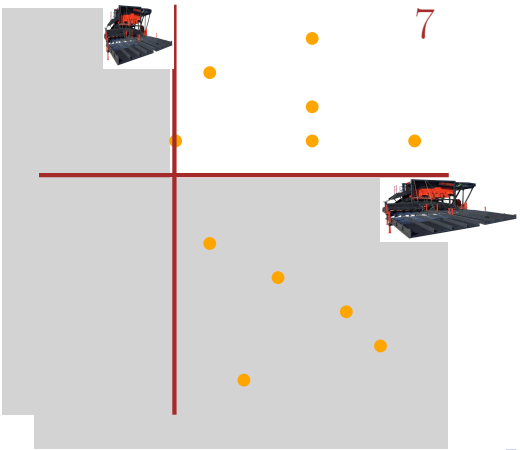

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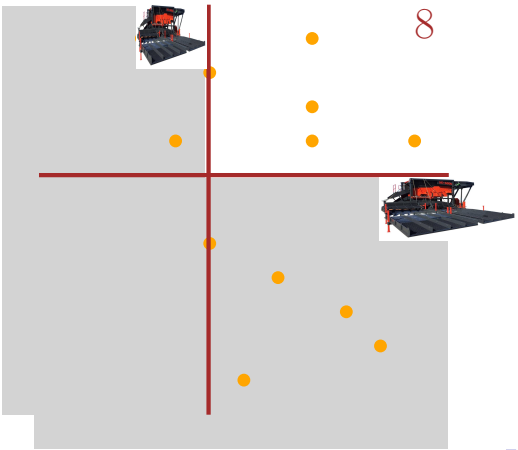

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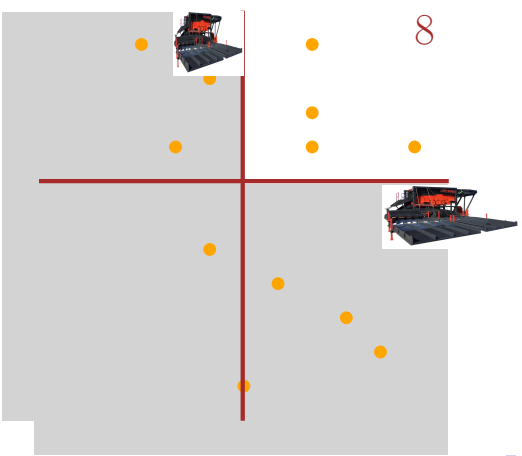

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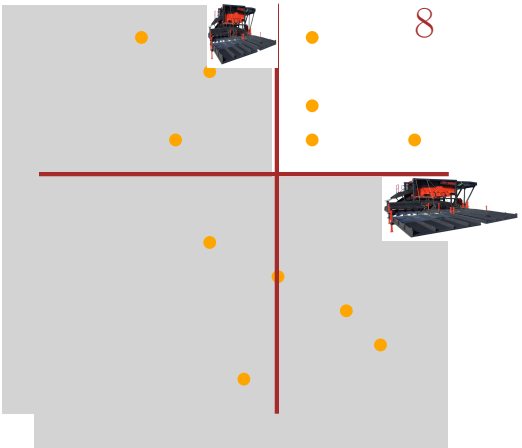

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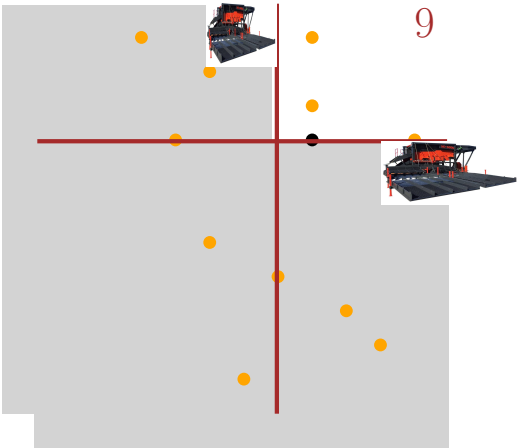

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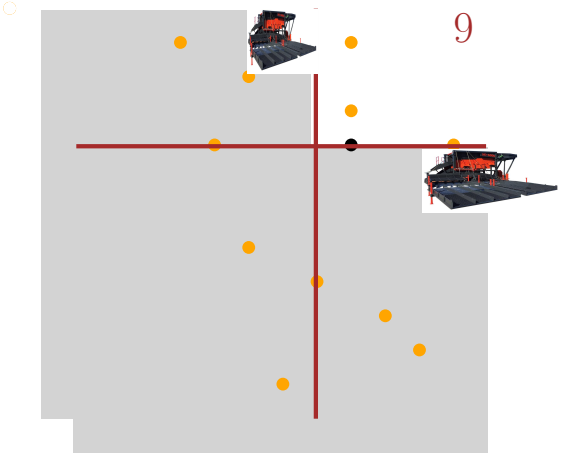

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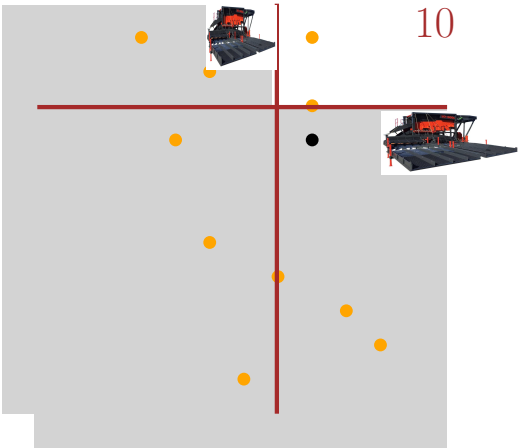

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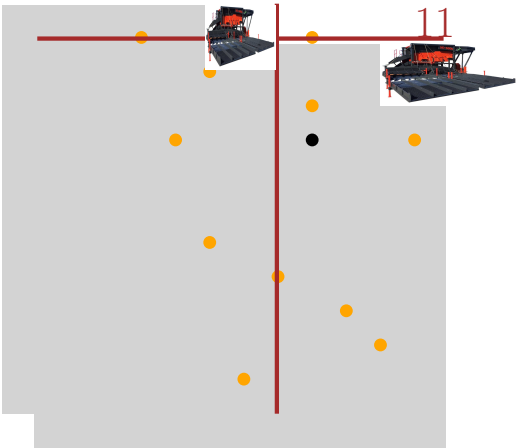

Greedy Approach

Greedy idea: At each step, advance the machine whose next position has *fewer* coins — preserving the richer direction for later.

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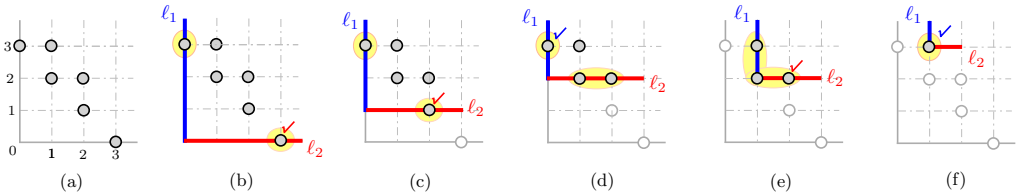

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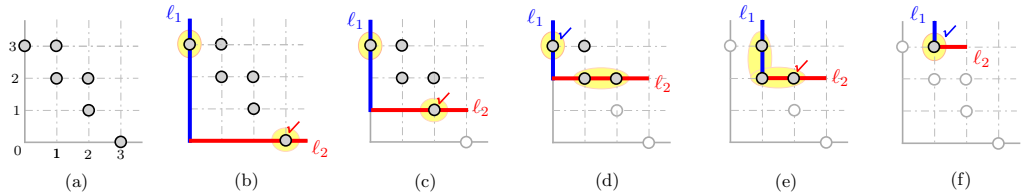

In practice, this heuristic performs well on random instances.

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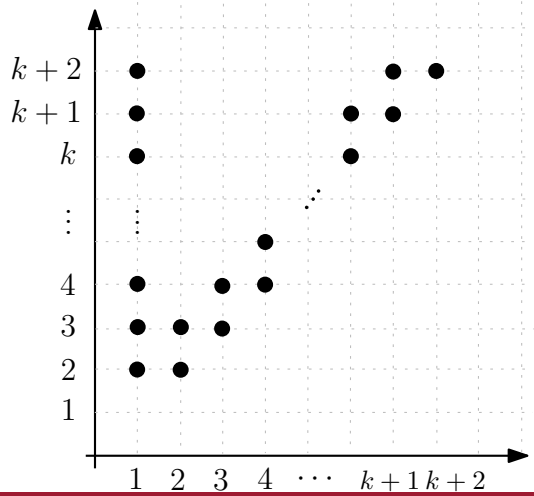

Dynamic Programming Solution

State: Let $\alpha(c, r)$ be the maximum coins collectible from the current position onward, when machine 1 is at **column** c and machine 2 is at **row** r.

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$$\alpha(c, r) = \max\Big(\underbrace{\alpha(c+1, r) + \mathbf{1}_c}_{\text{advance machine 1 (column)}} \quad , \quad \underbrace{\alpha(c, r+1) + \mathbf{1}_r}_{\text{advance machine 2 (row)}} \Big)$$

where $\mathbf{1}_c = 1$ if column c has a coin on row $r+$, $\mathbf{1}_r = 1$ if row r has a coin on column $c+$.

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Why DP?

The same state (c, r) is reachable via many different move sequences — **overlapping subproblems**. Storing $\alpha(c, r)$ avoids recomputation.

DP Solution: Filling the Table

- States (c, r) form an $n \times n$ table (column $\times$ row).
- Fill from the boundary (base cases at $c = n$ or $r = n$) toward $(0, 0)$.
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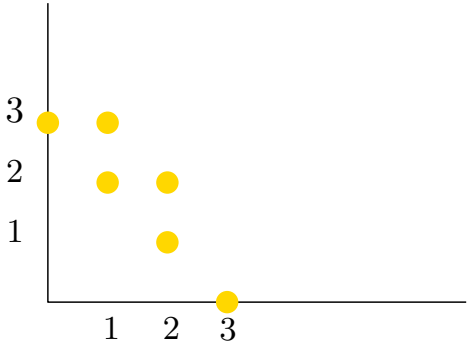

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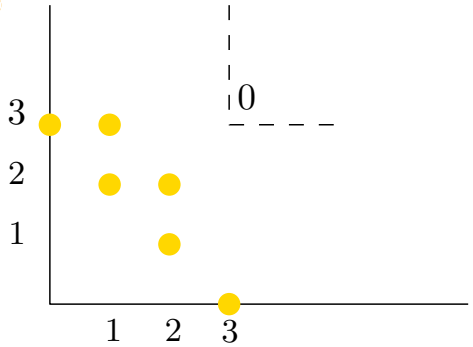

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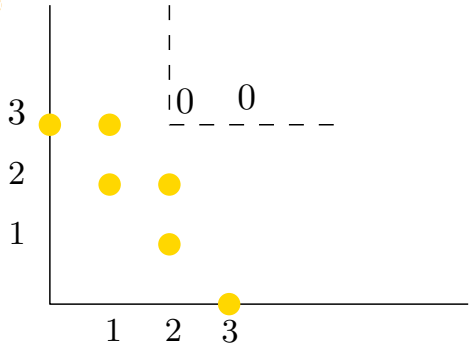

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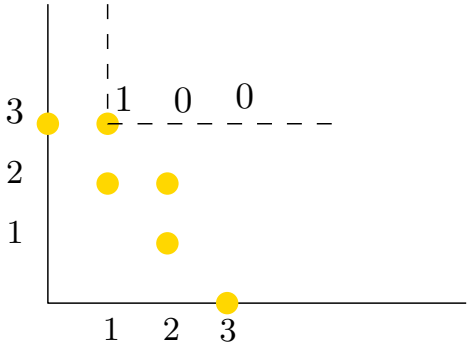

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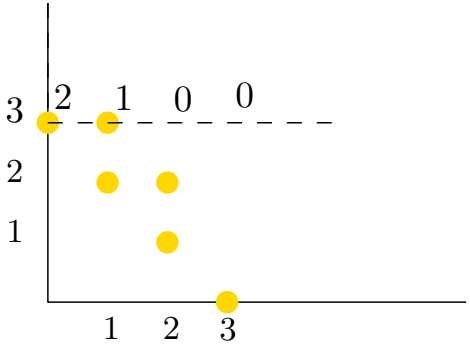

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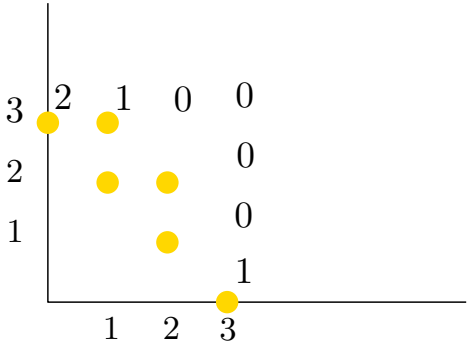

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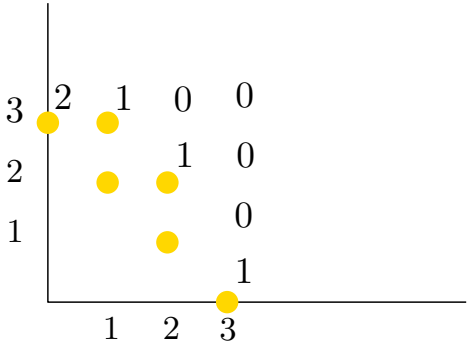

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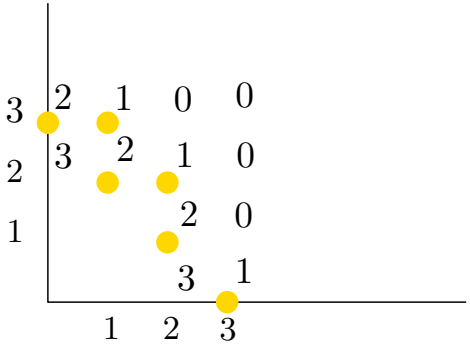

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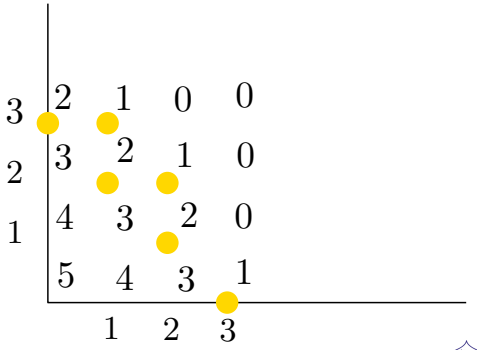

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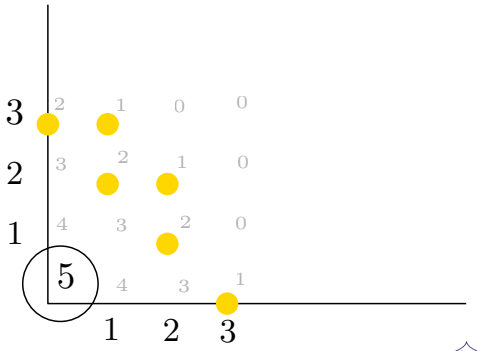

DP Solution: Filling the Table

- States (c, r) form an $n \times n$ table (column $\times$ row).
- Fill from the boundary (base cases at $c = n$ or $r = n$) toward $(0, 0)$.
- Each cell $\alpha(c, r)$ depends only on its right neighbor $\alpha(c + 1, r)$ and its top neighbor $\alpha(c, r + 1)$.

Time: $O(n^2)$ states, $O(1)$ work each $\Rightarrow O(n^2)$ total.